Chiral relations for entanglement in quantum self phase modulation

M. Gil de Oliveira* and A. Z. Khoury
Instituto de Física, Universidade Federal Fluminense, 24210-346 Niterói, RJ, Brazil
(Dated: July 22, 2026)

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2),\ \left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)\right] = \left[\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = 0 \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}. \tag{2}$$

We then have

$$\left[a(v_1), a(v_2)^\dagger\right] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r} = \langle v_1|v_2\rangle\,, \tag{3}$$

while $[a(v_1), a(v_2)] = \left[a(v_1)^\dagger, a(v_2)^\dagger\right] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}}\int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r}) + v(\mathbf{r})\mathcal{E}^\dagger(\mathbf{r})d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \tag{4}$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i\,\mathrm{Im}\,\langle v_1|v_2\rangle\,, \tag{5}$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2\Delta X(v_2)^2 \geq \frac{\left|\mathrm{Im}\,\langle v_1|v_2\rangle\right|^2}{4}. \tag{6}$$

One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i\,\mathrm{Re}\,\langle v_1|v_2\rangle \tag{7}$$

and

$$\Delta X(v_1)^2\Delta P(v_2)^2 \geq \frac{\left|\mathrm{Re}\,\langle v_1|v_2\rangle\right|^2}{4}. \tag{8}$$

We are interested in observables of the form

$$\Gamma(v_1, v_2) = \langle\{\Delta X(v_1), \Delta X(v_2)\}\rangle = \mathrm{Re}\left[\langle v_1|v_2\rangle + 2f(v_1, v_2) + 2h(v_1, v_2)\right], \tag{9}$$

* marcosgil@id.uff.br

where we have defined

$$\begin{aligned} f(v_1,v_2) &= \langle a(v_1)a(v_2)\rangle - \langle a(v_1)\rangle \langle a(v_2)\rangle \\ h(v_1,v_2) &= \left\langle a(v_2)^\dagger a(v_1)\right\rangle - \left\langle a(v_2)^\dagger\right\rangle \langle a(v_1)\rangle \end{aligned}. \tag{10}$$

In terms of the fields, the required quantities are written as

$$f(v_1,v_2) = \int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1,\mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad F(\mathbf{r}_1,\mathbf{r}_2) = \langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle - \langle \mathcal{E}(\mathbf{r}_1)\rangle \langle \mathcal{E}(\mathbf{r}_2)\rangle \tag{11}$$

$$h(v_1,v_2) = \int v_1^*(\mathbf{r}_1)v_2(\mathbf{r}_2)H(\mathbf{r}_1,\mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad H(\mathbf{r}_1,\mathbf{r}_2) = \left\langle \mathcal{E}^\dagger(\mathbf{r}_2,z)\mathcal{E}(\mathbf{r}_1,z)\right\rangle - \left\langle \mathcal{E}^\dagger(\mathbf{r}_2,z)\right\rangle \langle \mathcal{E}(\mathbf{r}_1,z)\rangle \tag{12}$$

This formulation is useful because f and g are normally ordered expectation values, which can be more easily calculated. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1,v_2+\lambda v_3) &= f(v_1,v_2) + \lambda^* f(v_1,v_3) \\ f(v_1,v_2) &= f(v_2,v_1) \\ h(v_1,v_2+\lambda v_3) &= h(v_1,v_2) + \lambda h(v_1,v_3) \\ h(v_1,v_2) &= h(v_2,v_1)^* \end{aligned} \tag{13}$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\gamma_{jk} = \langle \{Z_j, Z_k\}\rangle - 2\langle Z_j\rangle \langle Z_k\rangle$ where $\mathbf{Z} = (X(v), P(v))$. We observe that $h(v,v) \in \mathbb{R}$, so we get

$$\gamma = \begin{pmatrix} 1+2h(v,v) & 0 \\ 0 & 1+2h(v,v) \end{pmatrix} + 2\begin{pmatrix} \operatorname{Re} f(v,v) & -\operatorname{Im} f(v,v) \\ -\operatorname{Im} f(v,v) & -\operatorname{Re} f(v,v) \end{pmatrix}. \tag{14}$$

Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi} f(v,v)$. Therefore, we can diagonalize γ by choosing $\phi = [\arg f(v,v) \pm \pi]/2$. We then get eigenvalues

$$\lambda_\pm = 1 + 2h(v,v) \pm 2|f(v,v)|. \tag{15}$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_-, while the one rotated along the axis $\phi + \pi/2$ has maximum variance λ_+. Furthermore, we have squeezing if $|f(v,v)| > h(v,v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2, i.e., $\langle v_j|v_k\rangle = \delta_{jk}$. The Duan criterion [1] states that if

$$D = \frac{1}{2}\left\langle [\Delta X(v_1) + \Delta X(v_2)]^2\right\rangle + \frac{1}{2}\left\langle [\Delta P(v_1) - \Delta P(v_2)]^2\right\rangle < 1, \tag{16}$$

then the states are entangled.

Let us define $v_\pm = (v_1 \pm v_2)/\sqrt{2}$. We then have

$$D = \left\langle \Delta X(v_+)^2\right\rangle + \left\langle \Delta P(v_-)^2\right\rangle. \tag{17}$$

Applying (9) we obtain

$$\begin{aligned} D &= 1 + h(v_+,v_+) + h(v_-,v_-) + \operatorname{Re}\left[f(v_+,v_+) - f(v_-,v_-)\right] \\ &= 1 + h(v_1,v_1) + h(v_2,v_2) + 2\operatorname{Re} f(v_1,v_2) \end{aligned} \tag{18}$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_1 by an angle $\phi = \left[\arg f(v_1,v_2) \pm \pi\right]/2$. We get an optimal possible violation of

$$D_{\text{opt}} = 1 + h(v_1,v_1) + h(v_2,v_2) - 2|f(v_1,v_2)| \tag{19}$$

II. THE KERR SYSTEM

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z\mathcal{E}(\mathbf{r},z)+\nabla^2\mathcal{E}(\mathbf{r},z)=g\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z). \tag{20}$$

Let us decompose $\mathcal{E}=\mathcal{E}_0+\delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z\mathcal{E}_0(\mathbf{r},z)+\nabla^2\mathcal{E}_0(\mathbf{r},z)=0,\quad \mathcal{E}_0(\mathbf{r},0)=\mathcal{E}(\mathbf{r},0). \tag{21}$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r},z)+\nabla^2 u(\mathbf{r},z)=0$. Then, the anihilation operator corresponding to mode u is given by

$$a(u,z)=\int u^*(\mathbf{r},z)\mathcal{E}(\mathbf{r},z)d\mathbf{r}=a_0(u)+\delta a(u,z) \tag{22}$$

where

$$a_0(u)=\int u^*(\mathbf{r},z)\mathcal{E}_0(\mathbf{r},z)d\mathbf{r} \tag{23}$$

is independent of z (take a z derivative, use de equations of motion and integrate by parts) and

$$\delta a(u,z)=\int u^*(\mathbf{r},z)\delta\mathcal{E}(\mathbf{r},z)d\mathbf{r} \tag{24}$$

Our initial state will be a coherent one:

$$|\alpha\rangle=D(\alpha)\,|0\rangle=\exp\left(a(\alpha)^\dagger-a(\alpha)\right)|0\rangle\,. \tag{25}$$

Here, $\alpha(\mathbf{r})$ is a non normalized spatial function, with norm $\|\alpha\|$. This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r},z)\,|\alpha\rangle=\alpha(\mathbf{r},z)\,|\alpha,u\rangle \tag{26}$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},z)+\nabla^2\delta\mathcal{E}(\mathbf{r},z)=g\mathcal{E}^\dagger(\mathbf{r},z)\mathcal{E}^2(\mathbf{r},z). \tag{27}$$

Evaluating this at $z=0$ gives us

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r},0)=g\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{28}$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r},z)\approx-\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{29}$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r},z)\approx\mathcal{E}_0(\mathbf{r},0)-\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{30}$$

This expression will be the basis of our further developments.

B. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r},z)\rangle\approx\alpha(\mathbf{r},0)-\frac{igz}{2k}|\alpha(\mathbf{r},0)|^2\alpha(\mathbf{r},0). \tag{31}$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v \rangle = \langle v|\alpha\rangle - \frac{igz}{2k} \langle v, \alpha|\alpha, \alpha\rangle \tag{32}$$

In the above expression we have introduced the second spatial overlap

$$\langle v_1, v_2|u_1, u_2\rangle = \int v_1^*(\mathbf{r}, z) v_2^*(\mathbf{r}, z) u_1(\mathbf{r}, z) u_2(\mathbf{r}, z) d\mathbf{r}. \tag{33}$$

When $\alpha = \|\alpha\| u_{0l}$ is a Laguerre-Gaussian mode with radial order $p = 0$ and topological charge l, we have that

$$|u_{0l}(\mathbf{r}, z)| u_{0l}^2(\mathbf{r}, z) = \sum_{q=0}^{|l|} C_{ql} \overline{u}_{ql}(\mathbf{r}, z) \tag{34}$$

where $\overline{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v \rangle = \|\alpha\| \langle v|u_{0,l}\rangle - \frac{igz\|\alpha\|^3}{2k} \sum_{q=0}^{|l|} C_{ql} \langle v|\overline{u}_{ql}\rangle \tag{35}$$

In particular, when $v = \overline{u}_{ql}$, we have that

$$\langle a_{\overline{u}_{ql}} \rangle = \begin{cases} \|\alpha\| \langle \overline{u}_{ql}|u_{0,l}\rangle - \dfrac{igz\|\alpha\|^3 C_{ql}}{2k} & q \le |l| \\ \|\alpha\| \langle \overline{u}_{ql}|u_{0,l}\rangle & q > |l| \end{cases}. \tag{36}$$

C. Correlation functions

One of the two point field function is

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\delta\mathcal{E}(\mathbf{r}_2, z) + \delta\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) \tag{37}$$

where we have only kept terms linear in $\delta\mathcal{E}$.

Close to $z = 0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0) - \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0^\dagger(\mathbf{r}_2, 0)\mathcal{E}_0^2(\mathbf{r}_2, 0) - \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)^2\mathcal{E}_0(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0)\right] \tag{38}$$

Evaluating this at our initial state gives us

$$\begin{aligned} \left\langle \mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z)\right\rangle &\approx \alpha^*(\mathbf{r}_1, 0)\alpha(\mathbf{r}_2, 0)\left\{1 - \frac{igz}{2k}\left[|\alpha(\mathbf{r}_2, 0)|^2 - |\alpha(\mathbf{r}_1, 0)|^2\right]\right\} \\ &\approx \left\langle \mathcal{E}^\dagger(\mathbf{r}_1, z)\right\rangle \langle \mathcal{E}(\mathbf{r}_2, z)\rangle \end{aligned} \tag{39}$$

where the last approximate inequality comes from direct comparison with (31). In particular, this shows that, for any v_1 and v_2, we have $H(\mathbf{r}_1, \mathbf{r}_2) \approx 0$.

We can also use this to write the expectation value of the number operator:

$$\langle n(v)\rangle = \left\langle a^\dagger(v)a(v)\right\rangle \approx |\langle a(v)\rangle|^2 \approx |\langle v|\alpha\rangle|^2 + \frac{gz}{k}\,\mathrm{Im}\,\langle v|\alpha\rangle \langle \alpha, \alpha|\alpha, v\rangle \tag{40}$$

When both α and v are arbitrary Laguerre-Gaussian modes (any p, l, w), we have that $\mathrm{Im}\,\langle \alpha|u\rangle \langle \alpha, \alpha|\alpha, v\rangle = 0$, so there is no first order correction to the occupation number.

The other two point field function is

$$\begin{aligned} \mathcal{E}(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) &\approx \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0(\mathbf{r}_1, z)\delta\mathcal{E}(\mathbf{r}_2, z) + \delta\mathcal{E}(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k}\left[\mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0^\dagger(\mathbf{r}_2, z)\mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z)\right] \\ &= \mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_2, z)\mathcal{E}_0(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0^2(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \delta(\mathbf{r}_1 - \mathbf{r}_2)\mathcal{E}_0^2(\mathbf{r}_2, z)\right] \end{aligned} \tag{41}$$

The appearance of the delta function here is important, because now

$$\begin{aligned}\langle \mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle &\approx \alpha(\mathbf{r}_1,z)\alpha(\mathbf{r}_2,z)\left[1-\frac{igz}{2k}\left[|\alpha(\mathbf{r}_2,z)|^2+|\alpha(\mathbf{r}_1,z)|^2\right]\right]-\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z)\\ &\approx \langle \mathcal{E}(\mathbf{r}_1,z)\rangle\langle \mathcal{E}(\mathbf{r}_2,z)\rangle-\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z)\end{aligned} \tag{42}$$

Therefore, we may conclude that

$$F(\mathbf{r}_1,\mathbf{r}_2)\approx -\frac{igz}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)\alpha^2(\mathbf{r}_2,z), \tag{43}$$

$$f(v_1,v_2)\approx -\frac{igz}{2k}\langle v_1,v_2|\alpha,\alpha\rangle\,. \tag{44}$$

It will be useful to put this result in a dimensionless form. First, let us assume the input mode has a well define beam spot size w, and a Rayleigh range $z_R=kw^2/2$. By defining

$$Z=\frac{gz\|\alpha\|^2}{4z_R} \tag{45}$$

and $u(\mathbf{r})=w\alpha(\mathbf{r})/\|\alpha\|$ we get

$$f(v_1,v_2)\approx -iZ\langle v_1,v_2|u,u\rangle \tag{46}$$

and

$$\Gamma(v_1,v_2)=\mathrm{Re}\langle v_1|v_2\rangle+2Z\,\mathrm{Im}\langle v_1,v_2|\alpha,\alpha\rangle \tag{47}$$

This leads us to a minimum quadrature variance for mode v of

$$\lambda_-=1-|2Z\langle v,v|u,u\rangle| \tag{48}$$

and an optimal Duan parameter for modes v_1 and v_2 of

$$D_{\mathrm{opt}}=1-|2Z\langle v_1,v_2|u,u\rangle|. \tag{49}$$

The corresponding angles are $\phi_{\mathrm{squeezing}}=(-\pi/2\pm\pi+\arg Z\langle v,v|\alpha,\alpha\rangle)/2$ and $\phi_{\mathrm{duan}}=(-\pi/2\pm\pi+\arg Z\langle v_1,v_2|\alpha,\alpha\rangle)/2$

At this point, it becomes clear that the overlaps of the form $\langle v_1,v_2|\alpha,\alpha\rangle$ are the ones that dominate this sort of process.

Consider the case where all the involved modes are vortices: $u(\mathbf{r})=u_r(r)e^{il_u\phi}$ and similarly for the v functions. We have squeezing if $l_v=l_u$ and we have entanglement if $l_{v_1}+l_{v_2}=2l_u$. If all the radial functions are positive, as is the case for $LG_{0,0}$ and $LG_{0,\pm1}$, and α is positive, then the optimal angles are both $\mathrm{sign}(g)\pi/4$.

D. Selection Rules for two-mode entanglement in the first order treatment

Consider the case where the input mode is $\alpha(\mathbf{r})=\|\alpha\|LG_{0,l}(\mathbf{r})$, where

$$LG_{pl}(\mathbf{r};w)=(-1)^p\frac{\mathcal{N}_{pl}}{w}\left(\frac{2r^2}{w^2}\right)^{|l|/2}L_p^{|l|}\left(\frac{2r^2}{w^2}\right)e^{-r^2/w^2+il\phi} \tag{50}$$

is a Laguerre-Gaussian mode at the waist plane with radial order p, topological charge l and waist w. The normalization constant is

$$\mathcal{N}_{pl}=\sqrt{\frac{2}{\pi}}\sqrt{\frac{p!}{(p+|l|)!}} \tag{51}$$

Furthermore, let $v_2 \propto LG_{0,l_2}(\mathbf{r}; w_2)$ where w_2 is free and l_2 is only constrained by $l_2 \neq l$. Now, define $l_1 = 2l - l_2$ and $1/w_1^2 = 2/w^2 + 1/w_2^2$. We then have that

$$\begin{aligned} v_2^*(\mathbf{r})u^2(\mathbf{r}) &\propto \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2}{w_2}\left(\frac{2r^2}{w_2^2}\right)^{|l_2|/2}\left(\frac{2r^2}{w^2}\right)^{|l|} e^{-r^2/w_1^2 + il_1\phi} \\ &= \frac{\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2 w_1^{2|l|+|l_2|}}{w_2^{|l_2|+1}w^{2|l|}}\left(\frac{2r^2}{w_1^2}\right)^{|l_1|/2}\left(\frac{2r^2}{w_1^2}\right)^{p_{\max}} e^{-r^2/w_1^2 + il_1\phi} \end{aligned} \tag{52}$$

where we have defined

$$p_{\max} = \frac{1}{2}(2|l| + |l_2| - |2l - l_2|) = \begin{cases} 0, & ll_2 \leq 0 \\ \min(2|l|, |l_2|), & ll_2 > 0 \end{cases} \tag{53}$$

Then, by expanding the monomial that is raised to the power of $p_{\max}$ in terms of Laguerre polynomials, we get

$$v_2^*(\mathbf{r})u^2(\mathbf{r}) = \sum_{p=0}^{p_{\max}} C_p LG_{pl_1}(\mathbf{r}; w_1) \tag{54}$$

where

$$C_p = \frac{p_{\max}!\mathcal{N}_{0l_2}\mathcal{N}_{0l}^2 w_1^{2|l|+|l_2|+1}}{\mathcal{N}_{pl_1}w_2^{|l_2|+1}w^{2|l|}}\binom{p_{\max} + |l_1|}{p_{\max} - p} \tag{55}$$

Therefore, setting $v_1 = e^{i\phi}LG_{pl_1}(\mathbf{r}; w_1)$, which is orthogonal to v_2, and considering the two mode subsystem formed by v_1, v_2, we get an optimal Duan violation of

$$D_{\text{opt}} = \begin{cases} 1 - 2C_p Z, & p \leq p_{\max} \\ 1, & p > p_{\max} \end{cases} \tag{56}$$

Therefore, when $p \leq p_{\max}$, we have bipartite entanglement in this two-mode subsystem.

E. Multimode entanglement

Appendix A: Purely nonlinear approach

Let us make a modification to our equations of motion where

$$2ik\partial_z\mathcal{E}(\mathbf{r}, z) = \int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, z)\mathcal{E}(\mathbf{r} - \mathbf{s}, z)d\mathbf{s}\ \mathcal{E}(\mathbf{r}, z). \tag{A1}$$

In this form, the equation has a constant of motion $\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)$, which leads us to the exact solution

$$\begin{aligned} \mathcal{E}(\mathbf{r}, z) &= \exp\left[-\frac{iz}{2k}\int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, 0)\mathcal{E}(\mathbf{r} - \mathbf{s}, 0)d\mathbf{s}\right]\mathcal{E}(\mathbf{r}, 0) \\ &=: \exp\left[\int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k} - 1\right)\mathcal{E}^\dagger(\mathbf{r} - \mathbf{s}, 0)\mathcal{E}(\mathbf{r} - \mathbf{s}, 0)d\mathbf{s}\right] : \mathcal{E}(\mathbf{r}, 0) \end{aligned} \tag{A2}$$

In the last line, we used an important result known as the normal ordering theorem [2, 3]. This allows us to simply calculate the expectation value for the field in an initial coherent state:

$$\langle\mathcal{E}(\mathbf{r}, z)\rangle = \exp\left[\int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k} - 1\right)|\alpha(\mathbf{r} - \mathbf{s}, 0)|^2 d\mathbf{s}\right]u(\mathbf{r}, 0) \tag{A3}$$

We see that (31) is then a first order approximation of (A3) with $g(\mathbf{s}) = g_0\delta(\mathbf{s})$. Nonetheless, we see that we get singularities in the full expression when using this form of g. Let us then choose g as proportional to an approximation of the delta function. For definiteness, let us define

$$\delta_0(x, a) = \frac{\Theta(a/2 - |x|)}{a}, \tag{A4}$$

which approximates the delta distribution as $a \to 0$. Here, $\Theta(x)$ is the step function. Let us denote $\delta_0(\mathbf{r}, a) = \delta_0(x, a)\delta_0(y, a)$. We then choose $g(\mathbf{s}) = g_0\delta_0(\mathbf{s})$.

We will usually make the approximation that the support of the response function g is much smaller than the characteristic distances of the mode α. We can then make the approximation

$$\langle \mathcal{E}(\mathbf{r}, z)\rangle = \exp\left[G(z)|\alpha(\mathbf{r}, 0)|^2\right]\alpha(\mathbf{r}, 0) \tag{A5}$$

where

$$G(z) = \int_{\mathbb{R}^2}\left(e^{-izg(\mathbf{s})/2k} - 1\right)d\mathbf{s} = a^2\left(e^{-izg_0/2ka^2} - 1\right) \tag{A6}$$

Notice then that, in the general case, this is quite dependent on the value of a. Nonetheless, when $z|g_0|/2ka^2 \ll 1$, we get

$$G(z) \approx -\frac{izg_0}{2k}, \tag{A7}$$

which is independent of a.

Following the result of [3] we can calculate

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2) \approx &\alpha(\mathbf{r}_1, 0)\alpha(\mathbf{r}_2, 0)\exp\left[G(z)(|\alpha(\mathbf{r}_1)|^2 + |\alpha(\mathbf{r}_2)^2|)\right] \\ &\times\left[e^{-ig(\mathbf{r}_1-\mathbf{r}_2)z/2k + |\alpha(\mathbf{r}_1)|^2 G_1(\mathbf{r}_1-\mathbf{r}_2)} - 1\right] \end{aligned} \tag{A8}$$

where

$$\begin{aligned} G_1(\mathbf{r}, z) &= \frac{1}{2}\int_{\mathbb{R}^2}(e^{-izg(\mathbf{s}-\mathbf{r})} - 1)(e^{-izg(\mathbf{s})} - 1)d\mathbf{s} \\ &= \frac{a^4(e^{-ig_0z/2ka^2} - 1)^2}{2}\delta_1(\mathbf{r}, a) \\ &\approx -\frac{(g_0z/2k)^2}{2}\delta_1(\mathbf{r}, a) \end{aligned} \tag{A9}$$

where we have defined

$$\delta_1(x, a) = \frac{(a - |x|)\Theta(a - |x|)}{a^2} \tag{A10}$$

as another approximation of the delta function.

Notice, then, that, for modes with variation on a scale much larger than a, the second line of (A8) is essentially proportional to $\delta(\mathbf{r}_1 - \mathbf{r}_2)$, and we may write

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx \alpha(\mathbf{r}_1, z)^2\overline{F}(\mathbf{r}_1, a)\delta(\mathbf{r}_1 - \mathbf{r}_2) \tag{A11}$$

where

$$\alpha(\mathbf{r}, z) = \alpha(\mathbf{r}, 0)e^{-ig_0z|\alpha(\mathbf{r},0)|^2/2k} = \alpha(\mathbf{r}, 0)e^{-iZ|u(\mathbf{r},0)|^2/4} \tag{A12}$$

is the evolved field under the classical Kerr interaction and

$$\overline{F}(\mathbf{r}, a) = \int_{\mathbb{R}^2}\left\{\exp\left[-i\frac{g_0z}{2k}\delta_0(\mathbf{s}, a) - \frac{1}{2}\left(\frac{g_0z}{2k}\right)^2\delta_1(\mathbf{s}, a)|\alpha(\mathbf{r})|^2\right] - 1\right\}ds \tag{A13}$$

In general, this function presents a strong dependency on a. Nonetheless, if we also assume that $z^2|g_0|^2|\alpha(\mathbf{r})|^2/4k^2a^2 \ll 1$, then we may expand the exponential in a Taylor series and we get

$$\overline{F}(\mathbf{r}, a) = -\frac{g_0z}{2k}\left[i + \frac{g_0|\alpha(\mathbf{r})|^2z}{2k}\right], \tag{A14}$$

which does not depend on a. Substituting this result, we obtain

$$\begin{aligned} F(\mathbf{r}_1,\mathbf{r}_2) &\approx -\frac{g_0 z}{2k}\left[i+\frac{g_0|\alpha(\mathbf{r})|^2 z}{2k}\right]\alpha(\mathbf{r}_1,z)^2\delta(\mathbf{r}_1-\mathbf{r}_2) \\ &= -Z\left[i+Z|u(\mathbf{r})|^2\right]u(\mathbf{r}_1,z)^2\delta(\mathbf{r}_1-\mathbf{r}_2) \end{aligned} \tag{A15}$$

and

$$f(v_1,v_2) = -iZ\left\langle v_1,v_2|u(z),u(z)\right\rangle - Z^2\left\langle v_1,v_2,u(z)|u(z),u(z),u(z)\right\rangle \tag{A16}$$

Once again, following [3], we calculate

$$H(\mathbf{r}_1,\mathbf{r}_2) = \alpha^*(\mathbf{r}_1,z)\alpha(\mathbf{r}_2,z)(e^{|\alpha(\mathbf{r}_1)|^2 G_2(\mathbf{r}_1-\mathbf{r}_2)}-1) \tag{A17}$$

where

$$G_2(\mathbf{r}) = a^4\left|e^{ig_0 z/2ka^2}-1\right|^2\delta_1(\mathbf{r},a) \approx \left(\frac{g_0 z}{2k}\right)^2\delta_1(\mathbf{r},a) \tag{A18}$$

Therefore,

$$\begin{aligned} H(\mathbf{r}_1,\mathbf{r}_2) &\approx \left(\frac{g_0 z}{2k}\right)^2|\alpha(\mathbf{r}_1,z)|^4\delta(\mathbf{r}_1-\mathbf{r}_2) \\ &= Z^2|u(\mathbf{r}_1,z)|^4\delta(\mathbf{r}_1-\mathbf{r}_2) \end{aligned} \tag{A19}$$

and

$$h(v_1,v_2) = Z^2\left\langle v_1,u(z),u(z)|v_2,u(z),u(z)\right\rangle \tag{A20}$$

Notice that $h(v,v)\geq 0$, so it reduces both squeezing and the violation of Duan's inequality.

Appendix B: Numerical Considerations

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (20), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}},\tilde{z}) = \sqrt{\Delta A}\mathcal{E}(\mathbf{r},z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$\left[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}),\tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger\right] \approx \delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}, \tag{B1}$$

while the equation of motion reads

$$i\partial_{\tilde{z}}\tilde{\mathcal{E}}(\tilde{\mathbf{r}},\tilde{z}) = -\frac{1}{4}\tilde{\nabla}^2\tilde{\mathcal{E}}(\tilde{\mathbf{r}},\tilde{z}) + \frac{g}{4\Delta\tilde{A}}\tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}},\tilde{z})\tilde{\mathcal{E}}^2(\tilde{\mathbf{r}},\tilde{z}) \tag{B2}$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [4])

$$\begin{aligned} i\partial_{\tilde{z}}\psi_1(\tilde{\mathbf{r}},\tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_1(\tilde{\mathbf{r}},\tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_2(\tilde{\mathbf{r}},\tilde{z})\psi_1^2(\tilde{\mathbf{r}},\tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_1(\tilde{\mathbf{r}},\tilde{z})\xi_1(\tilde{\mathbf{r}},\tilde{z}) \\ -i\partial_{\tilde{z}}\psi_2(\tilde{\mathbf{r}},\tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_2(\tilde{\mathbf{r}},\tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_1(\tilde{\mathbf{r}},\tilde{z})\psi_2^2(\tilde{\mathbf{r}},\tilde{z}) + \sqrt{-\frac{ig}{4\Delta\tilde{A}}}\psi_2(\tilde{\mathbf{r}},\tilde{z})\xi_2(\tilde{\mathbf{r}},\tilde{z}) \end{aligned} \tag{B3}$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\left\langle\xi_j(\tilde{\mathbf{r}},\tilde{z})\xi_k(\tilde{\mathbf{r}}',\tilde{z}')\right\rangle = \delta_{jk}\delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}\delta_{\tilde{z},\tilde{z}'} \tag{B4}$$

For a coherent initial state in the mode u, we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^*(\tilde{\mathbf{r}}, 0)$.

[1] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).
[2] K. J. Blow, R. Loudon, S. J. D. Phoenix, and T. J. Shepherd, Continuum fields in quantum optics, Phys. Rev. A **42**, 4102 (1990).
[3] K. J. Blow, R. Loudon, and S. J. D. Phoenix, Exact solution for quantum self-phase modulation, J. Opt. Soc. Am. B **8**, 1750 (1991).
[4] P. Deuar and P. D. Drummond, Correlations in a bec collision: First-principles quantum dynamics with 150 000 atoms, Phys. Rev. Lett. **98**, 120402 (2007).